Turbid-Medium Imaging & Monte Carlo Light Transport — Project Plan

Sep 27, 2026 · @Shan

Abstract (draft)

We study light transport through a turbid milk phantom using a He-Ne laser (632.8 nm) and a CMOS camera. We ask whether a Monte Carlo (MC) model, calibrated only on object-free measurements, can quantitatively predict transillumination images of objects hidden in the medium.

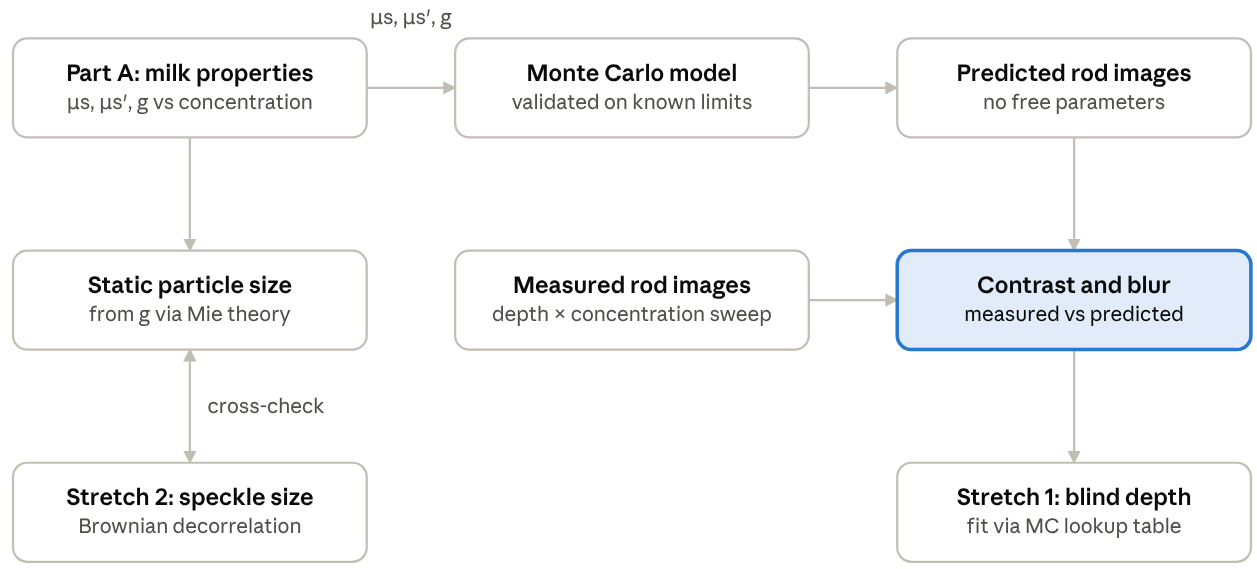
First, we characterise the medium. Collimated transmission gives the scattering coefficient μs, the lateral spread of a pencil beam gives the reduced scattering coefficient μs′, and their ratio yields the anisotropy g. We do this across concentrations for both skim and whole milk, and estimate an effective particle size from g using Mie theory.

Second, we write and validate a 3D photon-transport MC. With no free parameters, it predicts the shadow contrast and edge blur of an opaque rod as a function of depth and optical thickness, spanning the ballistic-to-diffuse transition. We compare these predictions with measurements.

Two extensions test the model and the medium further. An MC lookup table is used to infer the depth of a blindly placed rod. Speckle-contrast measurements of Brownian decorrelation give a dynamic particle size, compared with the static estimate from g.

1. Overview

The project asks one question: can an MC light-transport model, calibrated only on object-free measurements of milk, predict what a camera sees through that milk? Everything else either feeds that calibration or tests the model and the medium further.



project structure · 2 core parts, 2 stretch goals

Part A's parameters are the MC's only inputs. The rod comparison is the central result; each stretch goal hangs off one side of the project.

Tier	Part	Headline result
Core	A: medium characterisation	μs, μs′ and g vs concentration for skim and whole milk
Core	B: imaging prediction	Measured vs MC-predicted rod shadow contrast and edge blur, vs depth and optical thickness
Stretch	1: inversion	Rod depth inferred blind, with uncertainty, vs true depth
Stretch	2: speckle sizing	Particle radius from Brownian decorrelation, vs the Mie estimate from g

Minimum success is both core parts with error bars. Both stretch goals start in week 3 (Section 10). The abstract describes all four; the final report states how far each got.

2. Background theory

Four parameters describe the medium for our purposes: absorption μa, scattering μs, anisotropy g = ⟨cos θ⟩, and refractive index n. Everything we measure or simulate reduces to these.

2.1 Parameters and length scales

Quantity	Definition	Meaning
Extinction μt	μa + μs	Rate at which unscattered light is lost
Reduced scattering μs′	μs (1 − g)	Rate of effective direction randomisation
Scattering mean free path l	1 / μt	Average distance between scattering events
Transport mean free path l*	1 / (μa + μs′)	Distance after which direction is forgotten
Optical thickness τ	μt L	Tank length L in mean free paths

The useful measure of "how diffuse" a sample is for imaging is L / l*, not τ. Milk is strongly forward-scattering, so l* can be several times larger than l.

2.2 Ballistic light: Beer–Lambert

Unscattered light decays exponentially with path length. We reference every measurement to a water-filled tank, which cancels the wall Fresnel losses.

```
\frac{I_b}{I_{\text{water}}} = \exp\left(-\mu_t L\right)
```

At 633 nm absorption in dilute milk is expected to be tiny compared with scattering (approximate, to be confirmed), so μt ≈ μs.

2.3 Diffuse light and the similarity relation

When L ≫ l*, light has forgotten its initial direction and transport depends almost only on μs′, not on μs and g separately. This is why diffuse images alone cannot give g, and why we combine a ballistic measurement (μs) with a diffuse one (μs′):

```
g = 1 - \frac{\mu_s'}{\mu_s}
```

In this limit the diffusion approximation holds, with diffusion coefficient D = 1 / (3(μa + μs′)). We use it only as a validation check on the MC, not as our model.

2.4 Phase function

The MC samples scattering angles from the Henyey–Greenstein phase function, which is set by g alone. For a uniform random number ξ in (0, 1):

```
\cos\theta = \frac{1}{2g}\left[1 + g^2 - \left(\frac{1 - g^2}{1 - g + 2g\xi}\right)^2\right]
```

For g = 0 this is replaced by cos θ = 2ξ − 1. The azimuthal angle is uniform in (0, 2π).

2.5 Mie theory and particle size

For spheres of radius r in water, both g and the scattering cross-section depend on the size parameter x = 2π n_water r / λ and on the relative refractive index. Casein micelles (roughly 100–300 nm) give moderate g, fat globules (around 1 μm) give g close to 1 (approximate literature ranges, to verify). We compute g(r) with a Mie code, for example the miepython package, and invert our measured g for an effective radius.

2.6 Speckle contrast and Brownian motion

Coherent light scattered by moving particles forms a speckle pattern that fluctuates on a decorrelation time τc. A camera exposure of length T averages these fluctuations, lowering the speckle contrast K = σ_I / ⟨I⟩. For an exponentially decaying field correlation, with x = T / τc and β an optical factor set by speckle and pixel size:

```
K^2(T) = \beta \, \frac{e^{-2x} - 1 + 2x}{2x^2}
```

In the single-scattering regime τc = 1 / (D_B q²), where q = (4πn / λ) sin(θ/2) for scattering angle θ. The Stokes–Einstein relation then gives the hydrodynamic radius, with η the viscosity of water at the measured temperature T_s:

```
r = \frac{k_B T_s}{6 \pi \eta D_B}
```

2.7 Reading list

These are from memory, not a search, so check each exists and says what we need before citing.

- Wang, Jacques and Zheng (1995), MCML: the standard Monte Carlo code for layered tissue; our algorithm follows it.
- Bohren and Huffman, *Absorption and Scattering of Light by Small Particles*: Mie theory reference.
- Boas and Dunn (2010), review of laser speckle contrast imaging: the K(T) model above.
- Berne and Pecora, *Dynamic Light Scattering*: q-vector and Stokes–Einstein background.

3. Apparatus and sample preparation

One optical rail serves every part of the project; only the detector and the object change between measurements. Items marked [TBD] are placeholders to fill once we check the lab.

3.1 Equipment

Item	Role	Spec / status
He-Ne laser, 632.8 nm	Source for all parts	Power [TBD] mW, class [TBD]
Rectangular tank (glass or acrylic)	Holds the milk	Path length L [TBD]; 3–5 cm suggested
CMOS camera + lens with iris	Images exit face; speckle in Stretch 2	Model [TBD], bit depth [TBD], min exposure [TBD]
Photodiode + pinhole (or iris)	Collimated transmission (A1)	Pinhole placed ~1 m after tank
ND filters	Keep camera and diode below saturation	Set of several optical densities
Beam expander + ground-glass diffuser	Wide-field illumination (B, mode 2)	Rotating diffuser optional
Translation stage	Rod depth and/or beam scan	Availability [TBD]
Matte black rod	Test object	Diameter [TBD]; 2–5 mm suggested, spans full tank height

Item	Role	Spec / status
Thermometer	Water viscosity for Stretch 2	±0.5 °C is enough
Balance, pipettes, beakers	Dilution series	Standard lab kit

3.2 Sample preparation

1. Make a stock of skim milk powder at a fixed mass per volume of distilled water; record it. For the second medium use homogenised whole milk, which is more reproducible than whole-milk powder.
2. Define concentration c as the volume fraction of stock in the final sample. Make log-spaced dilutions so each decade gets several points.
3. Pour slowly down the tank wall to avoid bubbles, which scatter strongly. Stir gently before every measurement.
4. Fill the same tank with distilled water for a reference measurement at every session. This sets I_water and cancels wall losses.
5. Record temperature and time since mixing for every dataset.

3.3 Stability test (week 1)

Image the transmitted beam through one mid-range sample every minute for 20 minutes. If the signal drifts by more than a few percent, limit each sample's use to the stable window and remix. This test decides whether powder is usable at all.

3.4 Pilot run: choosing the concentration range (week 1)

Measure transmission at a coarse dilution series to find where $\tau = \mu t\, L$ reaches about 1 and about 5. Part A1 uses the low end; Part B needs L / l^* spanning roughly 1 to 10, which we estimate once $\mu s'$ is known.

3.5 Camera settings

- Fixed gain, fixed exposure within a dataset, auto-exposure and gamma off, raw or linear output.
- A dark frame at every exposure time used; subtract it from every image.
- Linearity check: image a steady beam through each ND filter and confirm signal scales with transmission.

- Never let the direct beam hit the sensor without ND filters.

4. Part A: medium characterisation

Part A produces the MC's only inputs: μs and μs′ per unit concentration, hence g, for skim and whole milk. It also gives a static particle size and tests where independent scattering breaks down.

A1. Collimated transmission → μs

1. Align laser → tank → pinhole about 1 m downstream → photodiode. The pinhole rejects scattered light; its acceptance half-angle is roughly pinhole radius / distance.
2. Measure I_water, then I_b for each low dilution, in random order to avoid drift correlating with c. Three repeats each.
3. Plot −ln(I_b / I_water) / L against c. In the independent-scattering regime this is a straight line through zero with slope ε = μt / c.
4. Leakage test: at the highest two concentrations, halve the pinhole size. If the result changes, forward-scattered light is leaking in; drop those points or shrink the pinhole.
5. Keep τ below about 3–4. Beyond that, scattered light dominates even through a small pinhole.

Output: specific extinction ε_s = μs / c (taking μa ≪ μs), with uncertainty from the fit. Curvature at high c, after the leakage test, signals dependent scattering.

A2. Exit-face spread → μs′

1. Use concentrations with L / l* around 3–10, estimated from A1 and a guessed g, then refined.
2. Send the pencil beam normally through the tank centre. Focus the camera on the exit face; record several exposures to cover the dynamic range of the spot.
3. Subtract dark frames, merge exposures, and azimuthally average about the beam centre to get a radial profile I(ρ).
4. Normalise the profile to its value at a fixed radius so the fit is sensitive to shape, not absolute power.
5. Fit with the MC (Section 5): run a grid of μs′ values with μa fixed small, and minimise χ^2 between measured and simulated profiles. Refine the grid near the minimum.
6. Repeat at 3–4 concentrations to check μs′ ∝ c.

Output: specific reduced scattering $\varepsilon_{s'} = \mu_{s'} / c$. Sensitivity to the assumed μ_a is tested by refitting with μ_a doubled.

A3. Anisotropy g

Compute $g = 1 - \varepsilon_{s'} / \varepsilon_s$. Using per-concentration coefficients lets A1 (dilute) and A2 (concentrated) combine, provided both stay in the linear regime. Propagate both fit uncertainties.

A4. Static particle size

With a Mie code, compute g as a function of radius for the relevant refractive indices. Read off the radius matching the measured g, with a band from the g uncertainty and from the refractive-index uncertainty. Milk is polydisperse, so this is an effective radius, weighted by how strongly each size scatters.

A5. Second medium

Repeat A1–A4 for homogenised whole milk. Fat globules are larger than casein micelles, so we expect a higher g and a different specific scattering; the size of that difference is a result in itself.

Part A outputs

Quantity	Skim	Whole
$\varepsilon_s = \mu_s / c$	[measured]	[measured]
$\varepsilon_{s'} = \mu_{s'} / c$	[measured]	[measured]
g	[derived]	[derived]
Effective radius (Mie)	[derived]	[derived]
Linear-regime limit c_{max}	[measured]	[measured]

5. Monte Carlo model

The MC is a weighted-photon random walk in 3D, following the MCML scheme, with a rectangular tank and an optional opaque cylinder. It must pass every validation test below before its predictions are compared with Part B.

5.1 Algorithm (per photon)

1. **Launch** at the entry face with weight w = 1. Pencil beam: fixed (x, y). Wide beam: x, y uniform over the illuminated area. Direction along +z.
2. **Step**: sample a path length $s = -\ln(\xi) / \mu t$.
3. **Boundaries**: if the step crosses a tank face, move to the face and apply Fresnel reflection or transmission (milk $n \approx 1.33$ to air; thin glass walls neglected at first, added later if needed). Transmitted photons are tallied and end.
4. **Object**: if the step intersects the cylinder (a quadratic in the x–z plane), the photon ends there as absorbed.
5. **Absorb**: $w \rightarrow w \cdot \mu s / \mu t$, and record the lost weight as absorbed.
6. **Scatter**: sample $\cos\theta$ from Henyey–Greenstein and φ uniformly, then rotate the direction vector.
7. **Roulette**: when $w < 10^{-4}$, keep the photon with probability 0.1 and multiply its weight by 10; otherwise end it.
8. Repeat from step 2.

5.2 Geometry and detector

- Tank: box of length L along z, lateral width W along x, height along y, matched to the real tank.
- Rod: infinite cylinder of radius a along y, centred at (x0, z0). The y direction then needs no binning.
- Detector: record exit position and direction of each transmitted photon. A photon counts toward the image only if its angle to the camera axis is below the lens acceptance half-angle, roughly arctan(aperture radius / working distance).
- Tallies: radial histogram $I(\rho)$ for Part A2; lateral histogram $I(x)$ for Part B; total transmitted weight for the scanned-beam mode.

The acceptance angle strongly affects predicted blur, so measure the real aperture and distance rather than guessing.

5.3 Implementation

- Python with NumPy, vectorised over batches of about 10^5 photons, or plain loops accelerated with Numba. Aim for 10^6–10^7 photons per configuration; large grids run overnight.

- One shared Git repository. Each run reads a config file (geometry, μs, μs′ or g, μa, source, detector, photon count, random seed) and writes results next to it.
- Statistical error: split each run into 10 batches and take the spread of batch results.

5.4 Validation tests

Test	Setup	Pass condition
Pure absorber	μs = 0	Transmission = exp(−μa L) × Fresnel factors
Ballistic fraction	Any μs, g	Unscattered fraction = exp(−μt L)
Energy conservation	Any	Reflected + transmitted + absorbed + object = 1
Similarity relation	g = 0 and g = 0.9 at equal μs′, L ≫ l*	Diffuse transmission agrees within MC noise
Diffusion limit	L / l* ⩾ 10	Total transmission close to diffusion theory for a slab
Convergence	Double photon count	Result changes by less than its statistical error
Object sanity	Rod in clear water	Sharp geometric shadow of width 2a

If time allows, also compare one case against a published MCML result.

6. Part B: imaging prediction

Part B measures the shadow of an opaque rod across depth and concentration and compares it with MC predictions made in advance, using only Part A's parameters. Agreement or explained disagreement is the project's central result.

6.1 Object and geometry

A matte black rod stands vertically across the full tank height, so the image is invariant along y and can be averaged into a 1D profile I(x). The rod slides along z (the beam axis) to set its depth z0. Depth, not rotation, is the experimental variable.

6.2 Measurement modes

Mode 1: scanned pencil beam (primary). Keep the rod fixed and translate the beam (or the tank) across x, for example in 0.5 mm steps over ±10 mm. At each position, record the total light reaching a fixed camera region around the exit point. This avoids speckle entirely and matches the MC's pencil-beam source directly.

Mode 2: wide-field (secondary). Expand the beam to cover the tank and pass it through the diffuser. Image the exit face and average rows along y to get I(x). Use exposures much longer than the speckle decorrelation time; check residual speckle by comparing short and long exposures.

6.3 Sweep

Variable	Values
Concentration	Water plus 4 levels spanning L / l* ≈ 1 to 10
Rod depth z0	5 positions: near entry, L/4, L/2, 3L/4, near exit
Mode	Scanned (all 25 configurations); wide-field (a subset of about 8)
Repeats	3 per configuration, sample remixed in between

Prediction files for every configuration are generated before the corresponding data are analysed. This keeps the comparison honest.

6.4 Comparison metrics

- **Shadow contrast** C = (I_bg − I_min) / I_bg, where I_bg is the level far from the rod.
- **Edge blur width** w: fit each edge with a step convolved with a Gaussian of width σ, or use the 10–90% rise distance.
- **Whole-profile agreement**: χ^2 per degree of freedom between measured and predicted I(x), each normalised to I_bg.

6.5 Plots for the report

1. Measured and predicted profiles overlaid, one panel per concentration at mid depth.
2. C against L / l*, one curve per depth, data points with error bars over MC lines.

3. w against depth at fixed concentration, showing blur growing as the rod moves away from the exit face.
4. Residuals (measured − predicted) for every configuration, to show systematic trends.

6.6 Complex-structure test

Two parallel rods separated by a gap slightly larger than the blur width at one concentration. The MC predicts whether they resolve as two minima or merge; the experiment checks it. Optionally, repeat with a sphere or an off-centre object if time remains.

7. Stretch 1: lookup-table inversion

We use the validated MC in reverse: given a measured rod profile, infer the rod's depth, and its diameter if unknown. The result is how precisely depth can be recovered as the medium becomes more diffuse.

7.1 Building the table

1. Pick two concentrations from Part B, one moderately and one strongly diffuse.
2. Run the MC over a depth grid (for example 9 depths from entry to exit face) and 2–3 rod diameters, in the scanned-beam geometry.
3. Store each predicted profile, normalised to I_bg. Interpolate linearly between neighbouring depths; check the interpolation against a few extra direct runs.

7.2 Blind trials

1. One team member places the rod at a depth (and, in some trials, one of the diameters), records it privately, and leaves.
2. The other two measure the profile and fit it to the table by minimising χ^2 over depth (and diameter).
3. Depth uncertainty comes from the range where χ^2 rises by 1 above its minimum, or from bootstrapping over repeat scans.
4. At least 5 trials per concentration; rotate who places the rod.

7.3 Expected physics: two degeneracies

- **Depth vs size.** In the diffuse regime a small shallow rod and a larger deeper one can cast similar shadows. A χ^2 map over (depth, diameter) shows this as a long valley rather than a point.
- **Front vs back.** A rod at depth z and one at L − z can give similar profiles when light is fully diffuse. We check this in the MC first.

If the front–back ambiguity appears, rotate the tank by 180° and take a second scan. Two opposite views break it. This is the one place in the project where multiple viewing angles carry real information.

7.4 Outputs

- Inferred against true depth, with error bars, for each concentration; the ideal is the line $y = x$.
- Depth resolution as a function of L / l^*.
- χ^2 maps illustrating the degeneracies, with and without the second view.

8. Stretch 2: speckle-contrast particle sizing

We measure how fast laser speckle from dilute milk decorrelates and convert that into a particle radius via Stokes–Einstein. Comparing it with the Mie radius from Part A gives two independent size estimates, one static and one dynamic.

8.1 Go / no-go check (week 1)

Find the camera's minimum exposure time. As a rough, unverified estimate, τ_c for particles of about 100 nm at $\theta = 90°$ is of order a millisecond, and longer for larger particles. We need a minimum exposure well below τ_c, ideally tens of microseconds.

If the camera is too slow, add glycerol to the water to raise the viscosity, which lengthens τ_c proportionally. Its viscosity and refractive index must then be known at the working temperature.

8.2 Setup

1. Use Part A1's calibration to choose a dilution with $\tau \lesssim 0.1$–0.2, so most detected light scattered once. A thinner cuvette helps here.

2. Observe scattered light at a fixed angle θ, for example 90° through the side wall. Measure θ carefully; q depends on it.
3. Set the lens iris so a speckle grain covers at least about 2 pixels. Check this with a static scatterer such as ground glass: the image autocorrelation width gives the grain size.
4. Record the sample temperature for every run.

8.3 Measurement

1. Sweep exposure time T logarithmically from the camera minimum to about 50 ms, with several points per decade.
2. At each T, keep the mean signal in the camera's linear range with ND filters; record around 20 frames.
3. Compute $K = \sigma_I / \langle I \rangle$ in small windows (for example 7 × 7 pixels) and average over windows and frames.
4. Correct for noise bias: subtract the dark-noise and shot-noise contributions to σ_I^2 before forming K.
5. Controls: the static ground glass should give $K \approx \sqrt{\beta}$ at every T; repeat one T with the table tapped to check vibration sensitivity.

8.4 Analysis

1. Fit $K^2(T)$ (Section 2.6) with β and τ_c free.
2. Compute q from n, λ and θ; then $D_B = 1 / (q^2 \tau_c)$.
3. Apply Stokes–Einstein with η of water (or of the glycerol mix) at the measured temperature to get r.
4. Propagate uncertainty from τ_c, θ and temperature.

8.5 Interpreting the comparison

The dynamic and static radii are weighted differently across milk's size distribution, so exact agreement is not expected. Agreement within a factor of about 2, or a disagreement we can explain by polydispersity or by fat versus casein, is a good result. Running skim and whole milk shows whether both methods see the same size difference.

9. Data analysis and uncertainty budget

Every reported number carries a 1σ uncertainty, and every fit reports χ^2 per degree of freedom. The table lists the error sources we expect and how each is controlled or estimated.

Source	Affects	Control or estimate
Dilution errors	ε_s, ε_s' slopes	Weigh stock; replicate two dilutions independently
Tank path length L	All μ values	Measure internal length with calipers
Wall Fresnel losses	Absolute transmission	Water reference each session
Scattered-light leakage (A1)	μt biased low	Pinhole-halving test; limit τ
Camera nonlinearity, dark signal	All profiles	Linearity test; dark frames at each exposure
Sample drift or settling	Time-dependent bias	Stability test; stir; stay in stable window
Residual speckle (mode 2)	Profile noise	Long exposures, diffuser, row averaging
Rod depth reading	z0 in Part B and Stretch 1	Stage scale; repeat placements
MC statistical noise	Predictions	Batch spread; more photons where needed
Detector acceptance angle	Predicted blur	Measure aperture and distance; vary in MC
Model assumptions (HG phase function, n, thin walls)	Systematic MC error	Vary each in the MC and report the spread
Temperature	Viscosity, hence Stretch 2 radius	Thermometer at each run
Scattering angle θ	q, hence Stretch 2 radius	Measure geometry; propagate

Working conventions

- Raw images are never overwritten. Each file name or log entry records sample, concentration, depth, mode, exposure and time.
- A shared lab log (one line per dataset) is kept alongside the code repository.
- Analysis scripts live in the same repository as the MC, so figures can be regenerated from raw data.
- Model-assumption effects are reported separately from statistical errors, since they are systematic.

10. Schedule and roles

Each person leads one workstream, and all three meet in the weekly lab session. Core parts finish by week 4; both stretch goals run alongside from week 3.

Core parts finish by week 4, leaving week 5 for stretch results and write-up

	P1: Monte Carlo lead	P2: optics and lab lead	P3: samples, analysis
Week 1	Core MC: slab, pencil beam Beer–Lambert, energy tests Set up shared Git repo	Assemble laser, tank, camera Camera linearity + darks Check min exposure (S2 go)	Milk stock, dilution plan Settling / stability test Pilot run: pick c range
Week 2	Fresnel walls, HG sampling Diffusion, similarity tests Exit-face and camera tally	A1: collimated transmission Skim and whole milk series Pinhole leakage test	A2: exit-face spot images Radial profile pipeline Fit μt vs concentration
Week 3	Add rod geometry to MC Fit μs′ from A2 profiles Predict Part B configs	Rod mount and depth stage B: scanned sweep begins S2: speckle optics setup	g and Mie radius (Part A) Profile metrics: C and w S2: first exposure sweep
Gate, end of week 3: does the MC match the A2 profiles? Yes → lookup table; no → debug first			
Week 4	Lookup table runs (S1) Bootstrap MC uncertainty Two-rod prediction	B: finish sweep, wide-field Two-rod experiment S1: blind depth trials	S2: full sweeps, fit τc Radius vs Mie estimate B: measured vs predicted
Week 5	S1: fit blind trials χ^2 maps, flipped-tank view Code clean-up for report	Retakes where data is weak Final figures	Error budget, final plots Draft report sections
All	End of week 5, all three: write-up, abstract update, final checks		

schedule · 5 weeks, 3 workstreams, 1 gate

The week-3 gate is the key checkpoint: the lookup table is only worth building once the MC reproduces the A2 profiles. The plan assumes about 6 hours per person per week (roughly 90 person-hours) plus overnight MC runs. Roles are a suggestion; swap them to match who prefers coding, optics or analysis.

11. Risks, safety and open items

The biggest schedule risk is the MC running late, because Part B and Stretch 1 both depend on it. Each risk below has an early warning sign and a fallback that still leaves a reportable result.

11.1 Risks and fallbacks

Risk	Early sign	Fallback
MC not validated by end of week 2	Validation tests still failing	P3 joins MC work; Part B uses scanned mode only; Stretch 1 shrinks to one concentration
Milk powder unstable	Drift in the week-1 stability test	Switch to homogenised liquid milk for both media
Scattered light leaks into A1	μt changes when the pinhole is halved	Smaller pinhole, longer distance, lower τ only
Residual speckle in wide-field images	Short and long exposures differ	Rotating diffuser, or rely on scanned mode
Camera too slow for Stretch 2	Minimum exposure not $\ll \tau c$	Glycerol–water mix to lengthen τc
MC and experiment disagree systematically	Same-sign residuals at all depths	Check acceptance angle, walls, phase function; report as a finding
Front–back depth ambiguity in Stretch 1	Flat χ^2 between z and L − z	Second scan with the tank rotated 180°
No translation stage	Lab check in week 1	Marked rod positions with a ruler; wide-field mode for Part B

11.2 Laser safety

- Check the laser's class on its label and follow the lab's rules for that class.
- Keep the beam horizontal and below eye level; block it at the end of the rail.
- Watch for reflections from the tank walls and cuvette faces; no watches or rings near the beam.
- ND filters go in front of the camera and photodiode before the beam is unblocked.
- Keep liquids away from the laser supply and the camera.

11.3 Open items

- [ ] Confirm whether the 6 hours per week is per person or per group
- [ ] Laser power and class
- [ ] Tank internal dimensions and wall material
- [ ] Camera model, bit depth, raw mode, minimum exposure
- [ ] Is a translation stage available, and for rod or beam?
- [ ] Rod material and diameter
- [ ] Skim milk powder brand; whole milk source
- [ ] Prof's sign-off on scope and the draft abstract